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# Dacheng Xu

## Education

2022.08–2027.05 (Expected) **Doctor of Philosophy in Physics**, *Columbia University*
Dissertation: Search for Light Dark Matter near the $^8$B Solar Neutrino Fog with XENONnT and a Krypton Background Measurement System at ppq Sensitivity
Advisor: Prof. Elena Aprile

2018.08–2022.08 **Bachelor of Engineering (Major: Engineering Physics)**, *Tsinghua University*
Dissertation: Probing Coherent Elastic Neutrino–Nucleus Scattering from Reactor Neutrinos with LXeTPC and HPGe Detectors
Advisor: Prof. Fei Gao & Prof. Litao Yang

## Selected Research Experience

### XENONnT Dark Matter Project

2022–2026 **Solar $^8$B CE$\nu$NS and Light Dark Matter Search**
Led solar neutrino analyses and developed signal and background models for light dark matter searches in the neutrino fog
- Led event selection, background modeling, and four-dimensional likelihood inference for XENONnT's first indication of solar $^8$B neutrino CE$\nu$NS and subsequent $3.3\sigma$ evidence from three science runs
- Corresponding author of two PRL papers (2024, 2026); main contributor to XENONnT's first light dark matter search in the neutrino fog (PRL, 2025)

2024–2026 **Analysis Coordinator of XENONnT**
Elected to coordinate collaboration-wide physics analyses, serving from October 2024 to February 2026
- Chaired analysis meetings, led analysis reviews, and coordinated working-group priorities and deadlines
- Coordinated shared data provisioning and analysis infrastructure; brought detector, calibration, background, and analysis teams together to reach common scientific decisions

2025–now **Krypton Background Measurement System**
Developing XTRACT, a krypton-in-xenon measurement system targeting parts-per-quadrillion (ppq) sensitivity for rare-event background characterization
- Contribute to cryogenic design, construction, and commissioning of a cold-trap separation and residual-gas analysis system
- Develop measurement automation and calibration to support krypton-background control for solar $pp$-neutrino and heavy WIMP searches

2024 **Convener of Computing and Analysis Tools Group**
Led the development of computing and analysis infrastructure supporting XENONnT physics analyses
- Developed and maintained HTC/HPC processing workflows involving 5 PB of I/O
- Designed a more stable data-reprocessing workflow adopted for subsequent XENONnT analyses

2023–2024 **Toy-MC Inference Framework**
Developed alea, an open-source statistical inference framework for toy-Monte-Carlo confidence interval construction
- Designed modular inference workflows integrated with HPC computing for large-scale toy-Monte-Carlo studies
- Adopted in XENONnT heavy WIMP, solar $^8$B neutrino, and $pp$-neutrino analyses, as well as by external collaborations

2022–2023 **GPU-based LXe Response Simulation and Inference**

Developed appletree, an open-source GPU framework for liquid-xenon detector response simulation and Bayesian inference

- Ported the simulation from CUDA/C++ to JAX/Python, improving usability and maintainability
- Enabled MCMC-based detector response fitting subsequently used in XENONnT dark matter and neutrino analyses, including sub-keV light and charge yield calibration with a $^{88}$YBe photoneutron source

### Jinping Neutrino Experiment (JNE)

2019–2021 **PMT Waveform Analysis for Neutrino Detection**

Led a study of photon timing and charge reconstruction for neutrino and dark matter experiments, published in JINST (first author, 2022)

- Developed waveform reconstruction algorithms and a common framework to benchmark convolutional neural networks, deconvolution, and Bayesian methods
- Improved photon counting and timing reconstruction, including charge resolution in low-occupancy regimes

### Gamma-Ray Integrated Detectors (GRID)

2018–2022 **CubeSat Gamma-ray Transient Events Analysis**

Developed a data processing pipeline for CubeSat gamma-ray burst (GRB) monitoring

- Built a pipeline subsequently used to detect over 40 GRBs, as of 2025
- Developed simulation tools for light curve analysis and localization algorithms

## Selected Honors and Awards

2026.10 **First Prize in the ARNPS Video Contest**, *Annual Review of Nuclear and Particle Science*

Led a team from three major xenon experiments to produce the educational video *Venturing into the Neutrino Fog*, explaining solar neutrino detection to early-career physicists. Contributed to concept development, scriptwriting, physics explanations, animation, and editing.
Prof. Michael E. Peskin: *"Our whole committee found this video very impressive."*

2022.03 **Grand Prize in the 17th "Challenge Cup" Competition**, *Ministry of Education, etc.*

Student Extracurricular Academic and Technological Works Competition
Project: GRID-02 satellite payload calibration and scientific results

2021.12 & 2019.12 **Dongchang He Prize**, *Tsinghua University*

Highest award for undergraduate of the Department of Engineering Physics, achieved twice

2020.12 **NAOC Fellowship**, *National Astronomy Observatory of China, Chinese Academy of Sciences*

Three winners per year in Tsinghua University, for students in the field of astronomy nationwide

2020.12 **BOEING Scholarship**, *Tsinghua University*

Excellent Comprehensive Scholarship of Tsinghua University

## Schools

2026.08 **Quantum Probes for a Quantum Universe (54th SLAC Summer Institute)**, *SLAC National Accelerator Laboratory, Menlo Park, US*

2026.07 **Qiskit Global Summer School 2026: A Decade on the Cloud (Quantum Excellence Badge)**, *IBM Quantum (online)*

Completed hands-on quantum-circuit programming labs using Python and Qiskit

## Teaching

2024 **Lecturer for XENONnT Computing Tutorial Series**, *Online*

Delivered biweekly tutorials on HTC/HPC workflows and analysis computing for XENONnT researchers

2022–2025 **Teaching Assistant**, *Columbia University*, Six Semesters

Intermediate Laboratory Work
Weekly lab sessions at E. K. A. advanced physics lab, covering topics in optics, quantum and particle physics

## Mentoring

2025 **Khushi Vandra**, *Columbia University*, NSF Research Experiences for Undergraduates (REU)
S1 pulse shape discrimination in liquid xenon detectors

## Internships

2024.07 & 2026.01 & 2026.07 **Visiting Researcher**, *Laboratori Nazionali del Gran Sasso*, Assergi, Italy
Operation and calibration shifts for XENONnT

2021 **Undergraduate Researcher**, *Columbia University*, Summer Undergraduate Research
Data analysis on the XENON detector, including several signal selection criteria
Supervisor: Prof. Elena Aprile

## Professional Services

2024–now **Journal Referee**
Journal of Instrumentation, Physical Review Applied

## Outreach Activities

2026.09–now **Math Enrichment Volunteer**, *Reading Team Math*, Harlem, New York
Help children build mathematical skills through weekly after-school games and play-based learning activities

2021–2026 **Open-source Projects Maintainer**, *Columbia University*
Maintained open-source analysis tools, including appletree and alea, and supported their use across collaborations; appointed as a straxen maintainer in 2025

2021.03 **Science Book Translator**
Translated John Haigh's *Probability: A Very Short Introduction* (Oxford University Press) into Chinese for a general audience

## Selected Publications

[1] E. Aprile et al. Probing the Solar $^{8}$B Neutrino Fog with XENONnT. *Phys. Rev. Lett.*, 137(9):091807, 2026. **Corresponding Author**.

*Main contribution:* Led the combined analysis of three science runs and coordinated manuscript preparation, establishing $3.3\sigma$ evidence for solar $^{8}$B neutrinos through coherent elastic neutrino–nucleus scattering and quantifying the impact of the neutrino fog on dark matter sensitivity.

[2] E. Aprile et al. WIMP Dark Matter Search Using a 3.1 Tonne-Year Exposure of the XENONnT Experiment. *Phys. Rev. Lett.*, 135(22):221003, 2025. **Main Contributor**.

*Main contribution:* Developed the simulation and statistical inference frameworks underpinning XENONnT's WIMP search, including fast signal simulation and dedicated modeling of accidental-coincidence backgrounds.

[3] E. Aprile et al. Low-energy nuclear recoil calibration of XENONnT with a $^{88}$YBe photoneutron source. *Phys. Rev. D*, 113(11):112017, 2026. **Main Contributor**.

*Main contribution:* Developed the fast detector-response simulation and Bayesian inference framework for multiple-scatter nuclear recoils, enabling sub-keV light and charge yield calibration for solar-neutrino and light-dark-matter searches.

[4] E. Aprile et al. First Search for Light Dark Matter in the Neutrino Fog with XENONnT. *Phys. Rev. Lett.*, 134(11):111802, 2025. **Main Contributor, Editors' Suggestion**.

*Main contribution:* Developed the signal and background models and four-dimensional likelihood framework that enabled XENONnT's first light dark matter search in the neutrino fog, including constraints on multiple dark matter interaction models.

[5] E. Aprile et al. First Indication of Solar $^{8}$B Neutrinos via Coherent Elastic Neutrino-Nucleus Scattering with XENONnT. *Phys. Rev. Lett.*, 133(19):191002, 2024. **Corresponding Author, Featured in Physics, Editors' Suggestion, APS Highlights of the Year**.

*Main contribution:* Led the analysis from low-threshold event selection and background modeling to four-dimensional likelihood inference, and coordinated manuscript preparation, establishing XENONnT's first indication of solar $^{8}$B neutrinos.

[6] C. Cai et al. Reactor neutrino liquid xenon coherent elastic scattering experiment. *Phys. Rev. D*, 110(7):072011, 2024. **Main Contributor**.

*Main contribution:* Developed the Geant4 detector simulation and sensitivity projections, informing detector and shielding design and quantifying sensitivity to the weak mixing angle and nonstandard neutrino interactions.

[7] E. Aprile et al. First Dark Matter Search with Nuclear Recoils from the XENONnT Experiment. *Phys. Rev. Lett.*, 131(4):041003, 2023. **Main Contributor, Featured in Physics, Editors' Suggestion**.

*Main contribution:* Developed the data-driven accidental-coincidence background model and a boosted-decision-tree classifier with 65% background rejection at 95% signal acceptance for XENONnT's first nuclear-recoil dark matter search.

[8] D. Xu et al. Towards the ultimate PMT waveform analysis for neutrino and dark matter experiments. *JINST*, 17(06):P06040, 2022. **First Author**.

*Main contribution:* Led algorithm development and established a common evaluation framework for PMT waveform reconstruction, systematically comparing convolutional neural networks, deconvolution, and Bayesian methods to improve photon counting and timing.

## Other Publications

[1] E. Aprile et al. The liquid-phase xenon purification system of the XENONnT dark matter experiment. arXiv:2611.00093, 2026.

[2] E. Aprile et al. First Measurement of Solar Neutrinos through Elastic Neutrino-Electron Scattering at the keV Scale. arXiv:2608.29450, 2026.

[3] E. Aprile et al. Search for Magnetic and Spin-Independent Inelastic Dark Matter with XENONnT. arXiv:2608.15149, 2026.

[4] E. Aprile et al. Direct Measurement of the $^{212}$Pb and $^{214}$Pb $\beta$ Decay Branching Ratios with the XENONnT Experiment. arXiv:2606.17920, 2026.

[5] E. Aprile et al. Enhancing Neutrinoless Double-Beta Decay Sensitivity of Liquid-Xenon Time Projection Chamber with Augmented Convolutional Neural Network. arXiv:2603.23549, 2026.

[6] E. Aprile et al. Light Dark Matter Search with 7.8 Tonne-Year of Ionization-Only Data in XENONnT. *Phys. Rev. Lett.*, 137(5):051003, 2026.

[7] E. Aprile et al. Constraints on Solar Reflected Dark Matter from XENON1T and XENONnT Data. *Phys. Rev. Lett.*, 137(12):121001, 2026.

[8] E. Aprile et al. Spectral measurement of the $^{214}$Bi $\beta$ decay to the $^{214}$Po ground state with the XENONnT Experiment. *Phys. Rev. C*, 113(4):044303, 2026.

[9] E. Aprile et al. Challenging Spontaneous Quantum Collapse with the XENONnT Dark Matter Detector. *Phys. Rev. Lett.*, 136(12):120201, 2026.

[10] E. Aprile et al. Radon Removal in XENONnT down to the Solar Neutrino Level. *Phys. Rev. X*, 15(3):031079, 2025.

[11] Kexin Liu, Dacheng Xu, and Fei Gao. The XENON dark matter and neutrino experiment. *Sci. Sin. Phys. Mech. Astro.*, 55(11):111004, 2025.

[12] E. Aprile et al. The neutron veto of the XENONnT experiment: results with demineralized water. *Eur. Phys. J. C*, 85(6):695, 2025.

[13] E. Aprile et al. Search for Light Dark Matter in Low-Energy Ionization Signals from XENONnT. *Phys. Rev. Lett.*, 134(16):161004, 2025.

[14] E. Aprile et al. XENONnT analysis: Signal reconstruction, calibration, and event selection. *Phys. Rev. D*, 111(6):062006, 2025.

[15] E. Aprile et al. Offline tagging of radon-induced backgrounds in XENON1T and applicability to other liquid xenon time projection chambers. *Phys. Rev. D*, 110(1):012011, 2024.

[16] E. Aprile et al. The XENONnT dark matter experiment. *Eur. Phys. J. C*, 84(8):784, 2024.

[17] E. Aprile et al. Design and performance of the field cage for the XENONnT experiment. *Eur. Phys. J. C*, 84(2):138, 2024.

[18] E. Aprile et al. Search for events in XENON1T associated with gravitational waves. *Phys. Rev. D*, 108(7):072015, 2023.

[19] E. Aprile et al. Searching for Heavy Dark Matter near the Planck Mass with XENON1T. *Phys. Rev. Lett.*, 130(26):261002, 2023.

[20] E. Aprile et al. Detector signal characterization with a Bayesian network in XENONnT. *Phys. Rev. D*, 108(1):012016, 2023.

[21] E. Aprile et al. The triggerless data acquisition system of the XENONnT experiment. *JINST*, 18(07):P07054, 2023.

[22] E. Aprile et al. Low-energy calibration of XENON1T with an internal $^{37}$Ar source. *Eur. Phys. J. C*, 83(6):542, 2023.

[23] E. Aprile et al. Effective field theory and inelastic dark matter results from XENON1T. *Phys. Rev. D*, 109(11):112017, 2024.

[24] E. Aprile et al. Search for New Physics in Electronic Recoil Data from XENONnT. *Phys. Rev. Lett.*, 129(16):161805, 2022.

[25] Xutao Zheng et al. In-orbit radiation damage characterization of SiPMs in the GRID-02 CubeSat detector. *Nucl. Instrum. Meth. A*, 1044:167510, 2022.

[26] Jia-Xing Wen et al. Compact CubeSat Gamma-ray detector for GRID mission. *Nucl. Sci. Tech.*, 32(9):99, 2021.

[27] Xiangyu Ivy Wang et al. GRB 210121A: A Typical Fireball Burst Detected by Two Small Missions. *Astrophys. J.*, 922(2):237, 2021.

[28] J. Wen et al. GRID: a student project to monitor the transient gamma-ray sky in the multi-messenger astronomy era. *Exper. Astron.*, 48(1):77–95, 2019.

## Talks

### Seminars

2026.09 **Venturing Into the Neutrino Fog: Solar Boron-8 Neutrinos and Light Dark Matter with XENONnT**, *Massachusetts Institute of Technology, Cambridge, US*, Laboratory for Nuclear Science (LNS) Lunchtime Seminars at MIT

2026.09 **Turning a Dark Matter Detector into a Neutrino Observatory: Latest Results from XENONnT**, *Yale University, New Haven, US*, Nuclear Particle Astrophysics (NPA) Seminar at Yale

2026.08 **Probing the Solar $^8$B Neutrino Fog with XENONnT**, *Lawrence Berkeley National Laboratory, Berkeley, US*, Institute for Nuclear and Particle Astrophysics (INPA) Seminar at LBNL

2026.07 **Probing the Solar $^8$B Neutrino Fog with XENONnT**, *University College London, London, UK*, HEP seminar at UCL

2026.03 **Latest solar neutrino and light WIMP search with XENONnT**, *Washington University in St. Louis, St. Louis, US*, WUCAP

2025.12 **Latest solar neutrino and WIMP search results from the XENONnT experiment**, *Kavli Institute for the Physics and Mathematics of the Universe, Kashiwa, Japan*

2025.12 **Latest solar neutrino and WIMP search results from the XENONnT experiment**, *Institute for Basic Science, Daejeon, Korea*

2025.04 **Venturing into the Neutrino Fog with XENONnT**, *SLAC National Accelerator Laboratory, Menlo Park, US*

2024.09 **A measurement of solar neutrino with XENONnT experiment**, *Stony Brook University, Stony Brook, US*

2024.09 **First Measurement of Solar $^8$B Neutrinos via Coherent Elastic Neutrino-Nucleus Scattering with XENONnT**, *Brookhaven National Laboratory, Upton, US*, Particle Physics Seminars at BNL

### Conferences

2026.07 **Talk: Probing the Solar Boron-8 Neutrino Fog with XENONnT**, *Max-Planck-Institut für Kernphysik, Heidelberg, Germany*
Magnificent CEvNS 2026

2026.06 **Poster: Measurement of Solar $^8$B Neutrino CE$\nu$NS with XENONnT**, *University of California, Irvine, Irvine, US*
Neutrino 2026: XXXII International Conference on Neutrino Physics and Astrophysics

2026.05 **Poster (People's Choice Award): XTRACT: The Measurement of Krypton in Xenon with ppq Sensitivity (public PDF)**, *TRIUMF, Vancouver, Canada*
XeSAT 2026 - International Workshop on Applications of Noble Gas Xenon to Science and Technology

2025.06 **Talk: A measurement of the Solar Boron-8 Neutrino Fog with XENONnT**, *Centro Brasileiro de Pesquisas Físicas, Rio de Janeiro, Brazil (online)*
Magnificent CEvNS 2025

2025.06 **Talk: Data processing in XENONnT dark matter project**, *University of Wisconsin-Madison, Madison, US (online)*
Throughput Computing 2025

2024.10 **Talk: First Measurement of Solar $^8$B Neutrinos via Coherent Elastic Neutrino-Nucleus Scattering with XENONnT**, *Blois, France*
Blois 2024: 35th Rencontres de Blois on "Particle Physics and Cosmology"

2024.07 **Talk: Search for Coherent Elastic Scattering of Solar $^8$B Neutrinos in the XENONnT**, *Shandong University, Qingdao, China*
The 14th Conference of High Energy Physics Branch of Chinese Physical Society

2021.08 **Poster: Accurate and Robust PMT Waveform Analysis**, *Shandong University, Qingdao, China (online)*
The 13th Conference of High Energy Physics Branch of Chinese Physical Society

2021.06 **Poster: Optimized PMT Waveform Analysis**, *University of Minnesota Twin Cities, Minneapolis, US (online)*
The 28th International Workshop on Weak Interactions and Neutrinos (WIN2021)

2020.08 **Talk: PMT waveform analysis in neutrino experiments**, *Lawrence Berkeley National Laboratory, Berkeley, US (online)*
Dark-matter and Neutrino Computation Explored (DANCE) Machine Learning Workshop 2020

## References

**Elena Aprile**, *Centennial Professor of Physics*, Columbia University
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